

## Lab 3 - Build Fabric Data Agent using Mirrored Azure SQL Database in Microsoft Fabric

### Introduction

Modern organizations need intelligent solutions that can analyze operational data and deliver business insights without requiring complex data movement or manual query development. Microsoft Fabric enables this by mirroring operational databases into OneLake and allowing AI-powered agents to query the data using natural language.

In this lab, you'll create an **Azure SQL Database** containing sample business data and configure **Azure SQL Mirroring** to replicate the data into Microsoft Fabric. Once the mirrored database is available, you'll create a **Fabric Data Agent** and connect it to the mirrored data source. The Data Agent uses natural language to understand user questions and generate insights without requiring SQL knowledge.

By the end of this lab, you'll be able to interact with operational data using conversational queries to explore product performance, customer information, and sales trends.

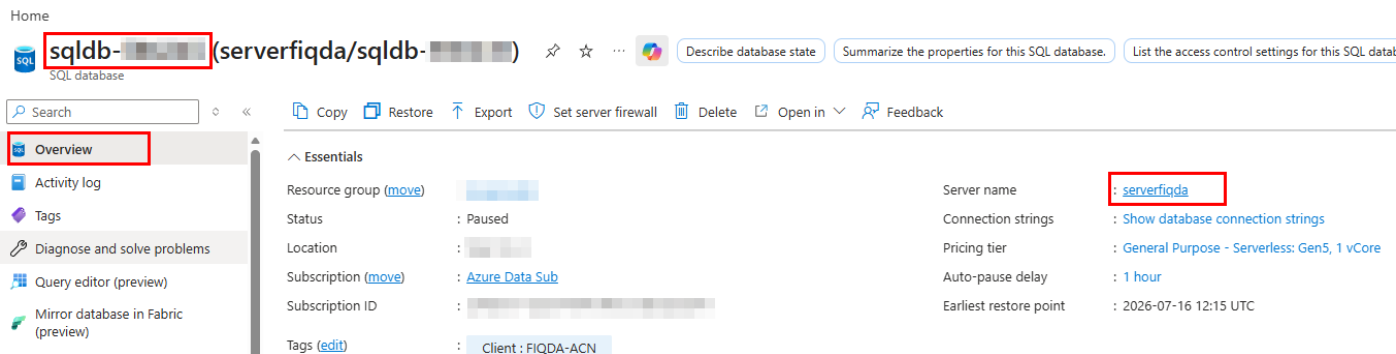
### Lab Objectives

In this lab, you will:

- Use the Created **Azure SQL Database** and populate it with sample business data.
- Configure **Azure SQL Mirroring** to replicate the database into Microsoft Fabric.
- Create a **Fabric Data Agent** and connect it to the mirrored database.
- **Validate** the Data Agent by asking natural language questions and analyzing the generated responses.

Task 1: Use the Created **Azure SQL Database** and populate it with sample business data.

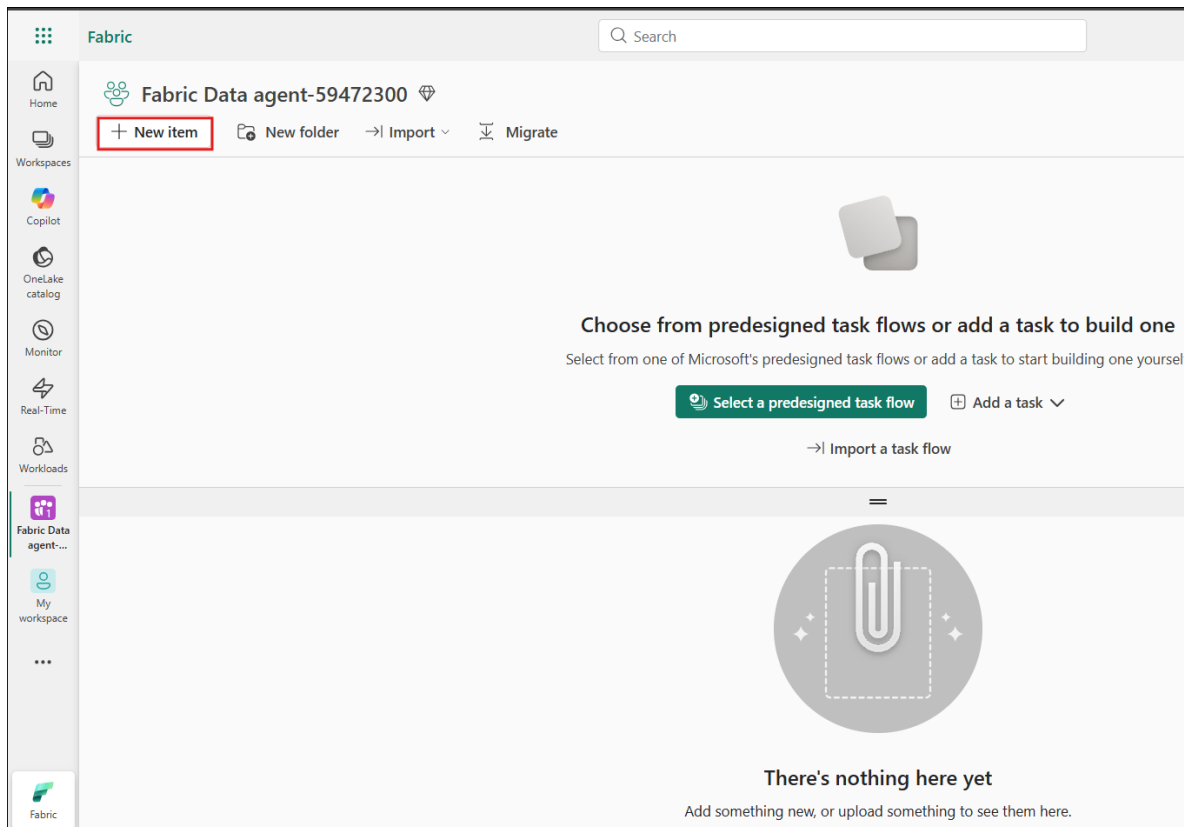
1. On your machine, open **Azure Portal** <https://portal.azure.com>
2. Search for Azure Database and select the database that is already created for you.
3. In SQL database page select **Query editor (preview) (1)** from the left navigation. In the **Query editor**, Use Microsoft Intra authentication, click on **Connect as**.
4. Expand the Sales LT Schema
5. On SQL Database **Overview** page, copy **Server name** and **SQL Database name (sqlldb- <randomsuffix>)** and paste them into the Notepad to use in the upcoming task.



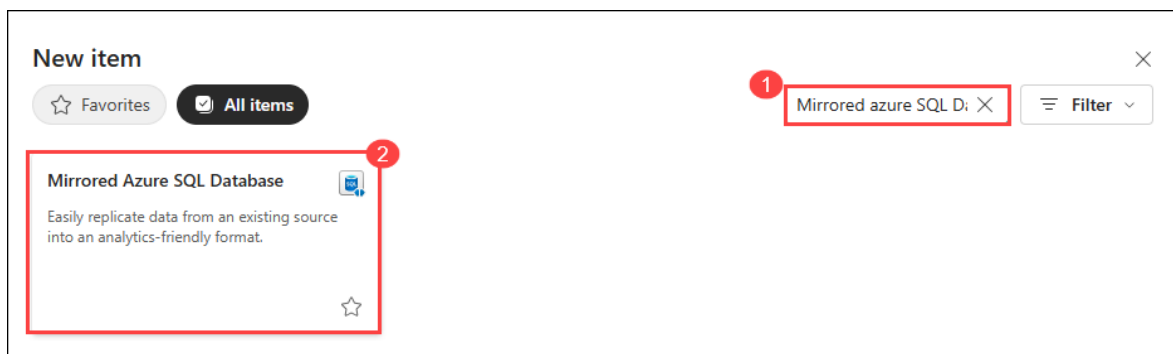
Task 2: Create a Solution to Mirror Data using Azure SQL Mirroring

In this task, you will connect the Azure SQL Database to Microsoft Fabric using Azure SQL Mirroring. You will select tables, create the mirrored database, and validate that the data has synced successfully.

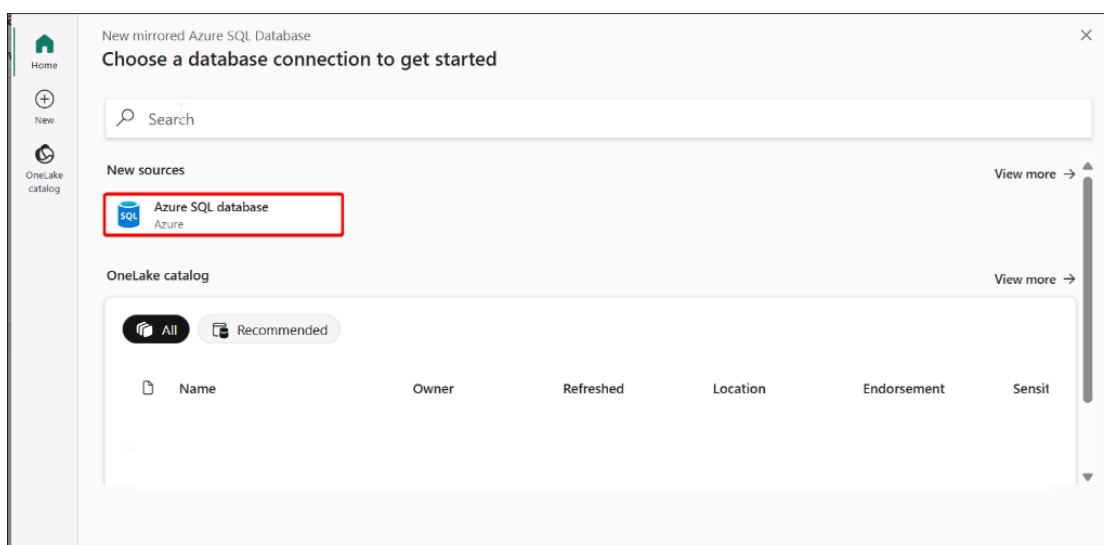
1. On your machine, open the **Microsoft Edge or any other browser**.
2. In the new tab, navigate to the **Microsoft Fabric** portal by copying and pasting the following URL into the address bar. <https://app.fabric.microsoft.com>
3. On the **Enter your email, we'll check if you need to create a new account** tab, you will see the login screen, in that enter the following email/username, and click on **Submit (2)**.
  - **Email/Username: (1)**
4. Next, provide your Temporary Access Password **(1)** and click on **Sign in (2)**:
  - **Tempromy Access Pass:**
5. If you see the pop-up Stay Signed in?, select **No**.
6. In the navigation bar, click on the **+ New item** button within the workspace **Fabric Data agent-**.



7. In the **Filter by keyword** search box, enter **Mirrored Azure SQL Database (1)** and select the **Mirrored Azure SQL Database (2)** item.



8. In the **Choose a database connection to get started** window, select **Azure SQL database**



9. In Connection settings tab enter the below detail and click on Connect button

| Setting             | Value                             |
|---------------------|-----------------------------------|
| Server              | <servername>.database.windows.net |
| Database            | Enter sqldatabase                 |
| Authentication Kind | Organization Account              |
| Click Switch        | You the provided credential       |
| Connect             | Click on Connect (5)              |

10. In the **Choose data** window, select **Select all (1)** and click on **Connect (2)** button

New mirrored Azure SQL Database

Choose data

**Data replication** Only valid columns are replicated to Fabric OneLake. You can set your security controls and policies from the SQL analytics endpoint. [Learn more](#)

Choose data

Search

12 out of 12 tables selected

☒ ☐ Select all **1**

☒ ☐ dbo.BuildVersion

☒ ☐ dbo.ErrorLog

☒ ☐ SalesLT.Address ⚠

☒ ☐ SalesLT.Customer ⚠

☒ ☐ SalesLT.Customer... ⚠

Select a table to see its preview

☒ **Automatically mirror future tables** Any new tables added to the source will be replicated to the destination.

Back **Connect **2****

11. In the Destination tab, click on **Create mirrored database**

New mirrored Azure SQL Database

Destination

**Azure SQL Database** [Learn more](#)

**Destination**

**Name \***

sqldatabase-

Back **Create mirrored database**

## 12. Click **Refresh** to update and view the latest changes.

Home

Stop replication | Configure replication | Manage OneLake data access

A SQL analytics endpoint for SQL querying and a default Power BI semantic model for reporting were created with this item.

Explorer

Search

Replication status

sqldatabase6789

Tables in OneLake

Details for sqldatabase6789

Mirrored database ID: fae6e071-71fc-4f19-8be1-...

Source: Azure SQL...

Source connection: sqlserver6758.database.windows.ne... 430cfd4a-4095-4dfa-aa2b-04c15cff1150

Create a semantic model

Create a new semantic model for Power BI reports using mirrored...

Create semantic model

Monitor replication

Here are the tables from the source. After replication is complete, the tables are then synced to the mirrored item's SQL analytics endpoint. You can also utilize workspace monitoring along with mirrored database operation logs to gain deeper insights and enable proactive monitoring.

Status

Running

Refresh

No data has landed

Once your data has landed, your data and replication status will appear here in OneLake.

Home

Stop replication | Configure replication | Manage OneLake data access

A SQL analytics endpoint for SQL querying and a default Power BI semantic model for reporting were created with this item.

Explorer

Search

Replication status

sqldatabase6789

Tables in OneLake

Details for sqldatabase6789

Mirrored database ID: fae6e071-71fc-4f19-8be1-...

Source: Azure SQL...

Source connection: sqlserver6758.database.windows.ne... 430cfd4a-4095-4dfa-aa2b-04c15cff1150

Create a semantic model

Create a new semantic model for Power BI reports using mirrored...

Create semantic model

Monitor replication

Here are the tables from the source. After replication is complete, the tables are then synced to the mirrored item's SQL analytics endpoint. You can also utilize workspace monitoring along with mirrored database operation logs to gain deeper insights and enable proactive monitoring. [Learn more](#)

Status

Running

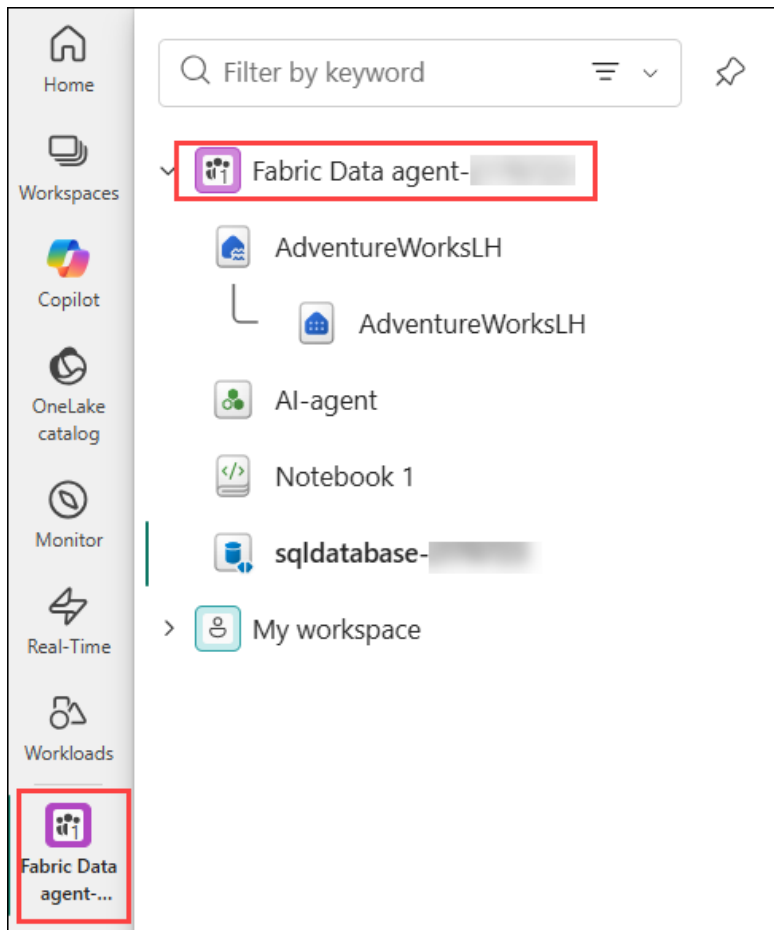
Refresh

| Name                                       | Status  | Rows replicated | Last completed |
|--------------------------------------------|---------|-----------------|----------------|
| [SalesLT].[ProductCategory]                | Running |                 | --             |
| [SalesLT].[ProductModelProductDescription] | Running |                 | --             |

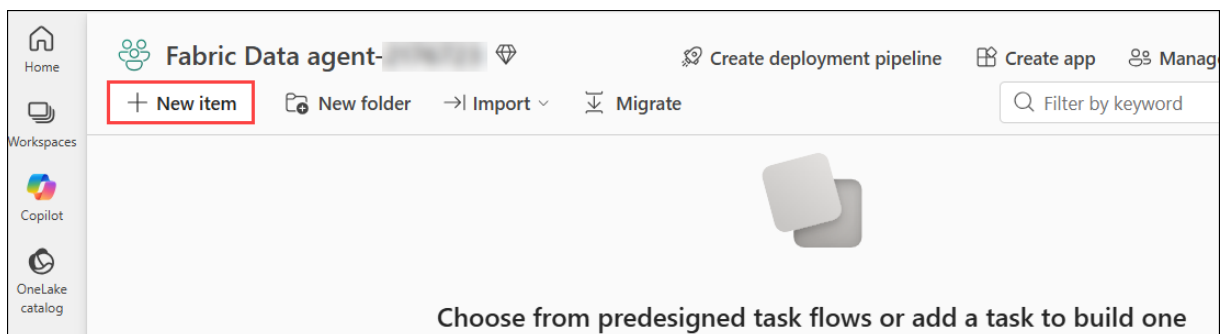
### Task 3: Create a Data Agent and Connect the Mirrored Database

In this task, you will create a new Fabric Data Agent and configure it to use the mirrored Azure SQL Database as its data source. This agent will respond to natural language prompts using the mirrored data.

1. In the left-sided navigation menu, navigate and click on **Fabric Data agent-**.



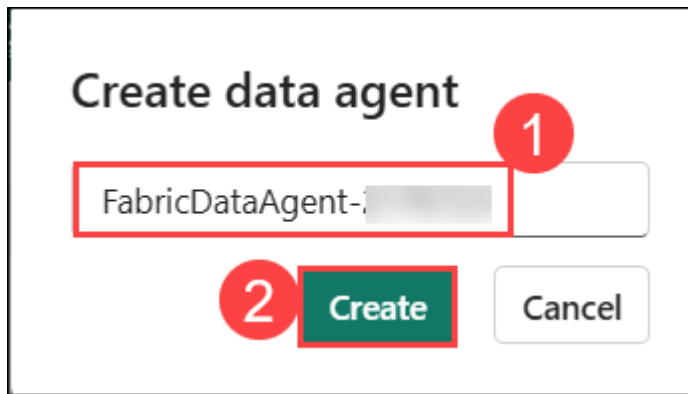
2. In the **Fabric Data agent-** workspace home page, select **+ New item**.



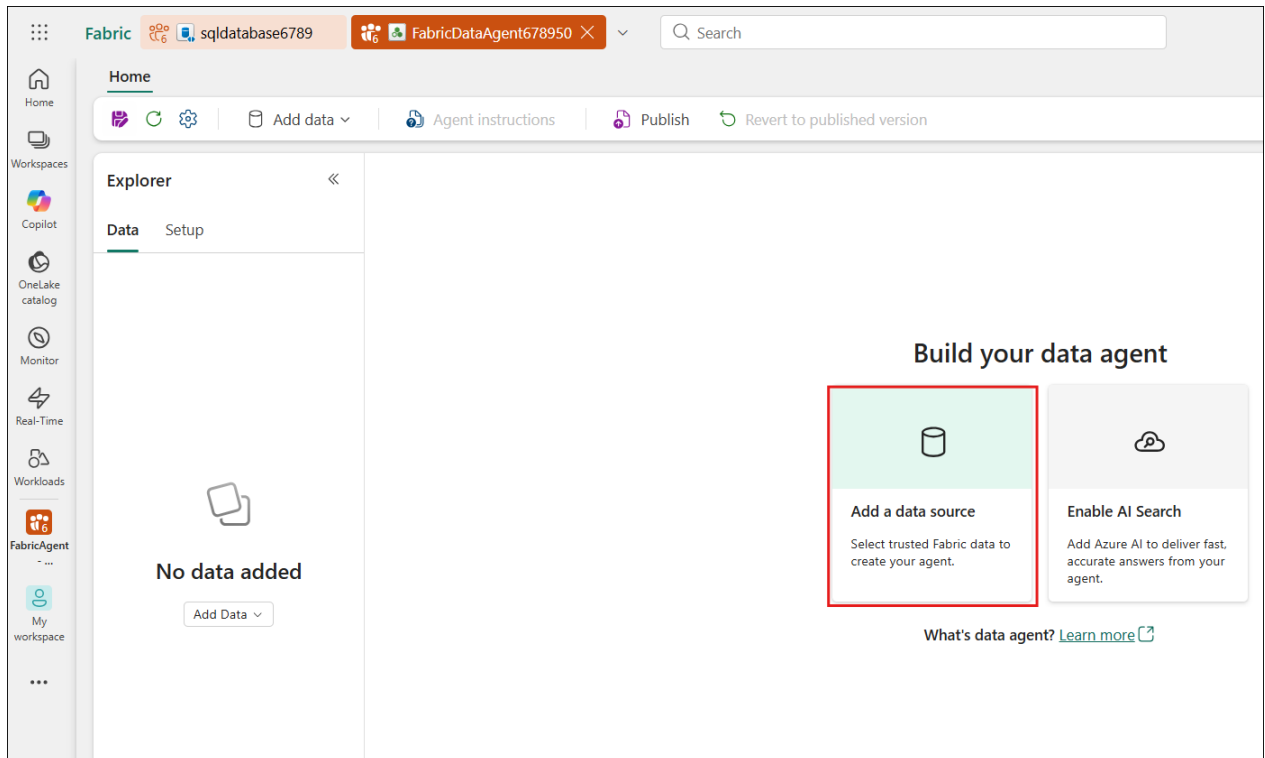
3. In New item pane, search for **Data agent (1)** and select the **Data agent (2)**



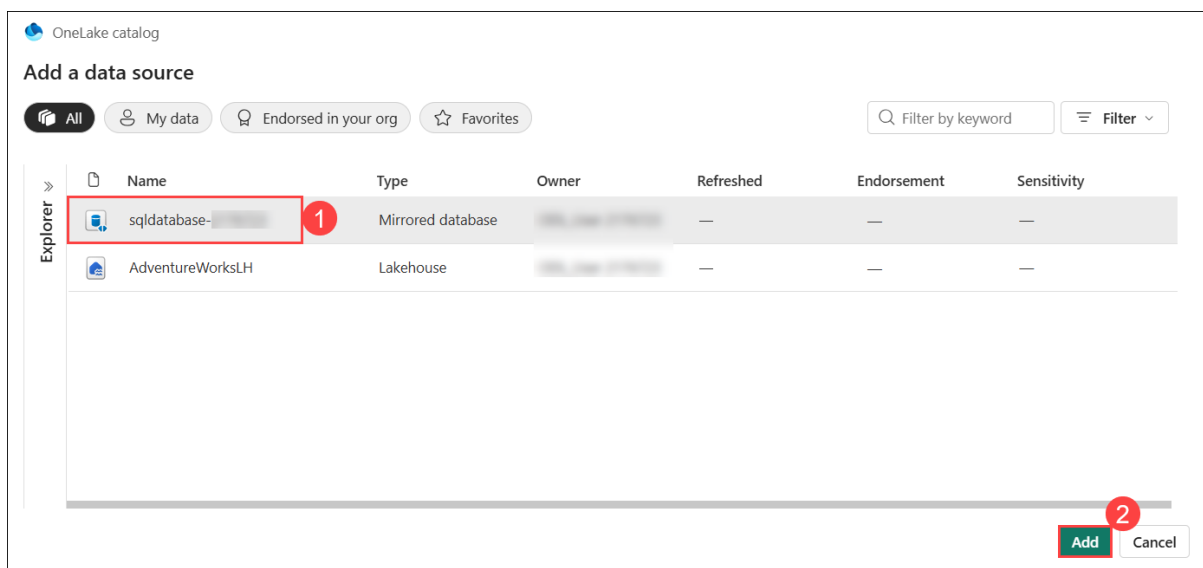
4. Enter **FabricDataAgent-** (1) as the Data agent name and select **Create (2)**.

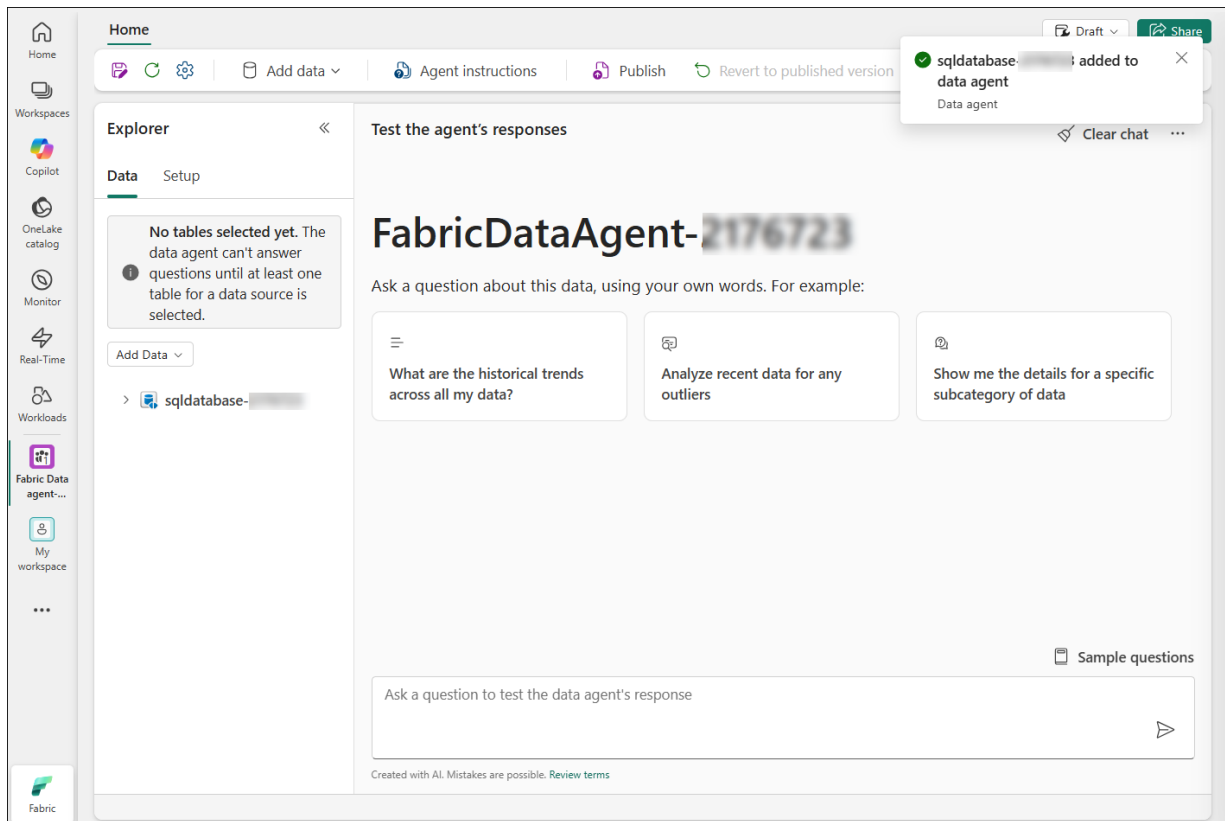


5. Select **Add data source** to configure a new data source.



6. Select **sqldatabase- (1) Mirrored database resource**, then click **Add (2)**.





Task 4: Test the agent and Validate the agent responses using sample analytical questions.

In this task, you will test the Data Agent by asking analytical questions like:

- Which product categories generate the highest sales?
- List products with high list price but low sales volume.
- Which cities have the highest number of customers?

This validates the agent's ability to understand and respond to business queries.

1 Expand the **sqldatabase- (1)**, navigate through **Schemas (2) → SalesLT (3) → Tables (4)**, and **select all (5)** the tables.

1. Enter the following **question (6)** in the query box:

Which product categories generate the highest sales?

2. Review the response **(7)**, which lists the top product categories along with their total sales amounts (e.g., Category ID 7, 6, and 5 as the highest).



**Explorer**

Data Setup

Add Data

sqldatabase- 1

Schemas 2

dbo

SalesLT 3

Tables 4 5

Address

Customer

CustomerAddress

Product

ProductCategory

ProductDescription

ProductModel

ProductModelProductDescri...

SalesOrderDetail

SalesOrderHeader

**Test the agent's responses**

Clear chat

Which product categories generate the highest sales? 6

The product categories that generate the highest sales are:

1. Product Category ID 7 - with a total sales amount of 221,081.50
2. Product Category ID 6 - with a total sales amount of 185,510.28
3. Product Category ID 5 - with a total sales amount of 173,065.85

These categories lead in terms of total sales generated. If you want to know the actual names of these product categories, let me know!

1 step completed 7

Response time: 17 sec

Sample questions

Ask a question to test the data agent's response

Created with AI. Mistakes are possible. Review terms

3. To test the agent, run the application and enter the sample questions to verify the responses.

List products with high list price but low sales volume.

**Home**

Draft

Add data Agent instructions Publish Revert to published version

**Explorer**

Data Setup

Add Data

sqldatabase6789

dbo

BuildVersion

ErrorLog

SalesLT

Address

Customer

CustomerAddress

Product

ProductCategory

ProductDescription

ProductModel

ProductModelProduct...

SalesOrderDetail

SalesOrderHeader

**Test the agent's responses**

Clear

Which product categories generate the highest sales?

The product categories that generate the highest sales are:

1. Product Category ID 7
2. Product Category ID 6
3. Product Category ID 5

These categories have the highest total sales amounts among all product categories. If you need the actual names or more details about these categories, let me know!

1 step completed

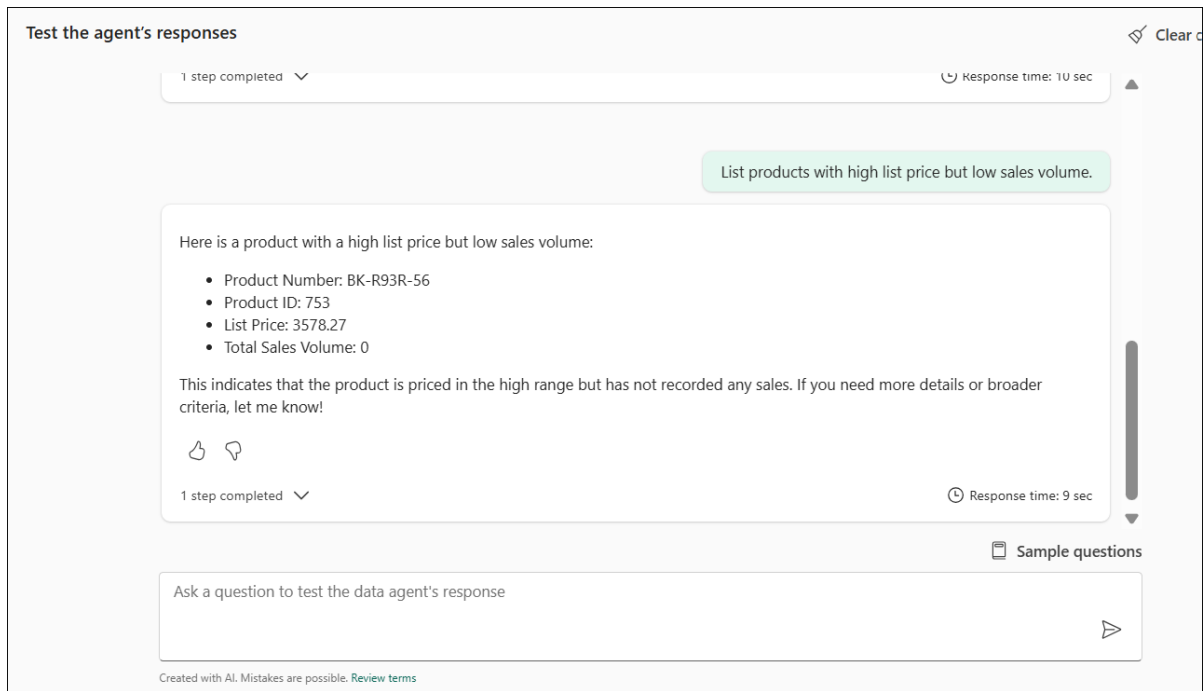
Response time: 9 sec

Sample questions

List products with high list price but low sales volume 1

2

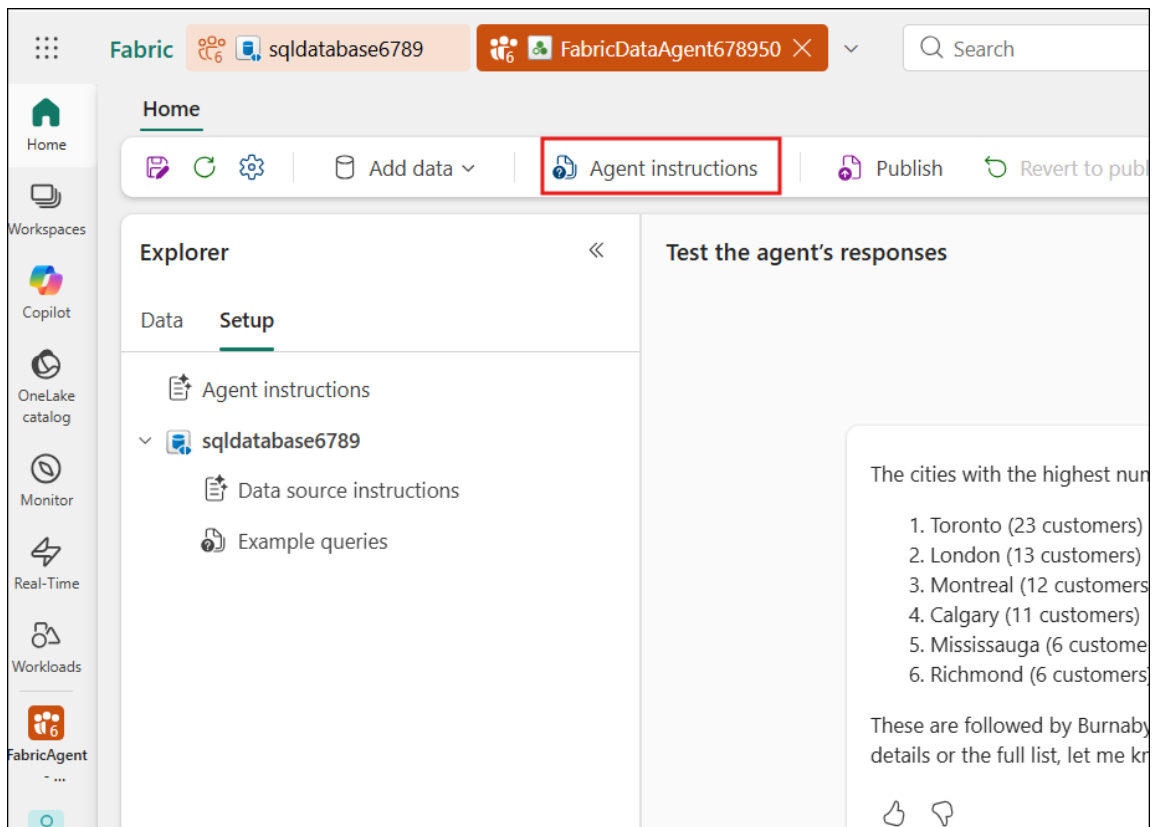
Created with AI. Mistakes are possible. Review terms



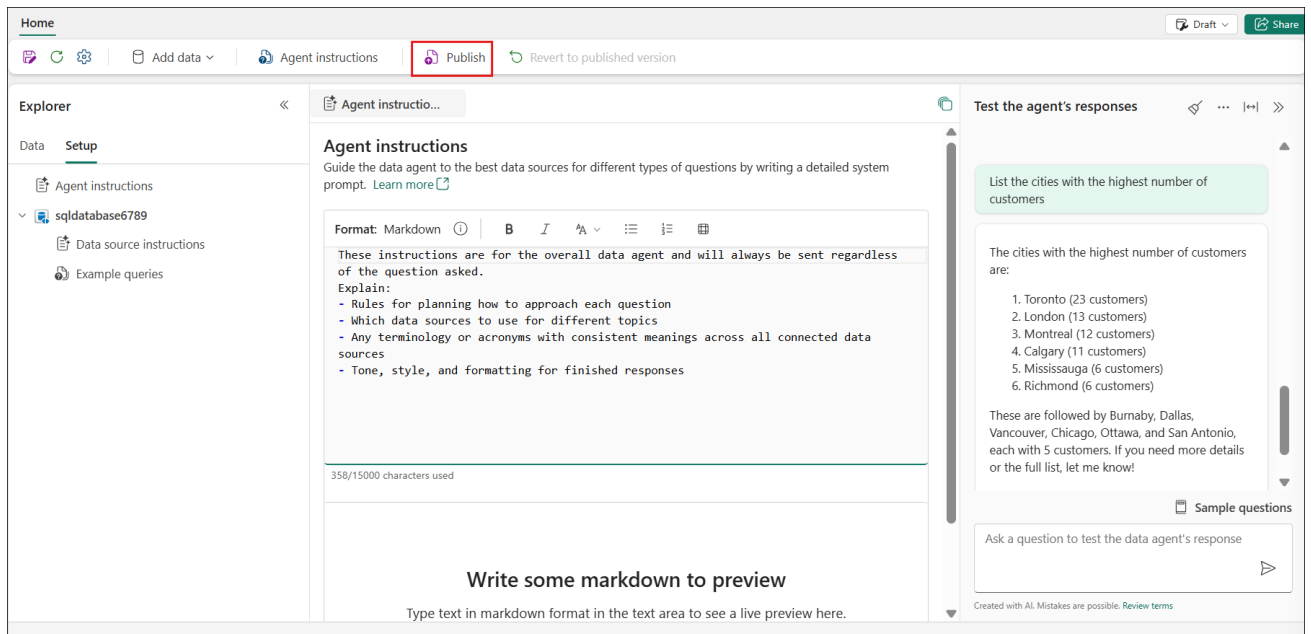
4. Similarly, enter the following **question** in the query box:

List the cities with the highest number of customers

5. Click **Agent instructions** from top menu.



6. Click Publish from the top menu and select **Publish**.



## Publish data agent

Name

FabricDataAgent-2157188

Description of purpose and capabilities

Explain what kinds of questions this agent can answer and what data it can access.

Also publish to the Agent Store in Microsoft 365 Copilot

Available immediately for personal use and sharing

Off

2 Publish Cancel

